

SYDNEY TECHNICAL HIGH SCHOOL



TRIAL HIGHER SCHOOL CERTIFICATE
2001

MATHEMATICS 4 UNIT

*Time allowed - Three hours
(Plus 5 minutes reading time)*

Name: Class:

This test paper must be handed in with your answers

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	Question 8	TOTAL

DIRECTIONS TO CANDIDATES

- Attempt ALL questions
- ALL questions are of equal value
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are printed at the back of this test booklet.
- *Board-approved* calculators may be used.
- *Each* question is to be started on a new page clearly marked Question 1, Question 2, etc.. Each page must show your name.
- You may ask for more paper if you need it.

An academically selective school for boys

QUESTION 1:

(a) Find

1 (i) $\int \frac{dx}{x^2 + 2x + 5}$

2 (ii) $\int_0^1 \frac{dx}{(x+1)\sqrt{x+1}}$

2 (b) Prove that $\sec x = \frac{\sec x \tan x + \sec^2 x}{\sec x + \tan x}$

and hence find $\int \sec x dx$

2 (c) (i) Find the exact value of $\int_0^1 x e^{-x} dx$

4 (ii) Find $\int \frac{5 dx}{(x+1)(x^2 + 4)}$

2 (d) Find $\int_k^1 \frac{dx}{x(x+1)}$ and hence prove that

2
$$\sum_{k=1}^n \int_k^1 \frac{dx}{x(x+1)} = \log_e(n+1) - n \log_e 2$$

QUESTION 2:

(a) If $z = 3-4i$ find

6 (i) $\bar{z}$ (ii) $|z|$ (iii) $\arg z$ (iv) $\arg(iz)$ (v) $\sqrt{z}$

2 (b) The complex number $z = x+iy$ is such that $|z-i| = \operatorname{Im}(z)$

Find, and describe geometrically, the locus of the point P representing z

4 (c) Sketch the locus on the Argand Diagram of the point Z representing the complex number z where $|z-2i|=1$

What is the least value of $\arg z$?

3 (d) A is the point representing the complex number $z = 2+3i$, while B represents the complex number iz .

The point C is such that AOBC is a square (where O is the origin)
Find the co-ordinates of C.

QUESTION 3:

- 3 (a) If one root of the polynomial equation $x^3 + ax^2 + bx + c = 0$ is the sum of the other two roots, show that

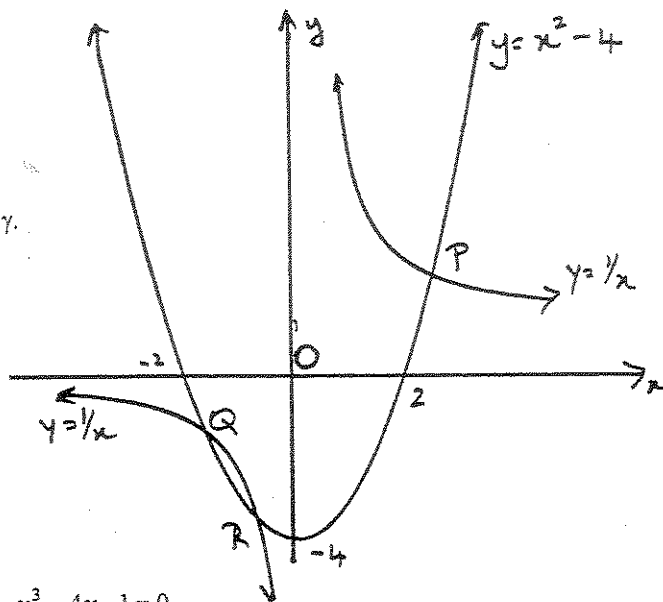
$$a^3 - 4ab + 8c = 0$$

- 3 (b) The polynomial $P(x) = x^3 + ax^2 + bx + 6$ where a and b are real numbers, has a zero of $1-i$.

Find a and b and express $P(x)$ as the product of two polynomials with real coefficients.

(c)

The curves $y = \frac{1}{x}$ and $y = x^2 - 4$ intersect at points P, Q, R as shown. P, Q and R have x-values α , β and γ . O is the origin.



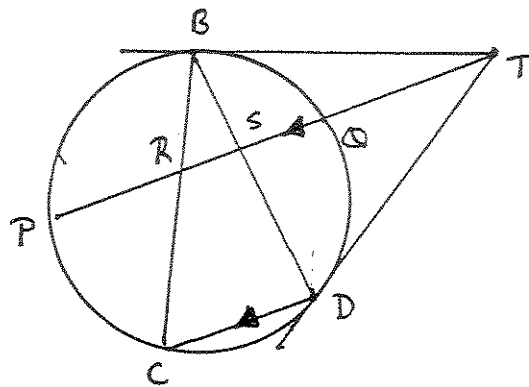
- 1 (i) Show that α , β and γ are roots of $x^3 - 4x - 1 = 0$
- 2 (ii) Find a polynomial with numerical coefficients with roots α^2 , β^2 , and γ^2
- 3 (iii) Find an expression for $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$
- 3 (iv) Hence find the value of $OP^2 + OQ^2 + OR^2$

QUESTION 4:

(a) Given the hyperbola $9x^2 - 16y^2 = 144$ find

- | | |
|---|---------------------------------------|
| 1 | (i) the length of the major axis |
| 1 | (ii) the eccentricity |
| 1 | (iii) the co-ordinates of the foci |
| 1 | (iv) the equations of the directrices |
| 1 | (v) the equations of the asymptotes |

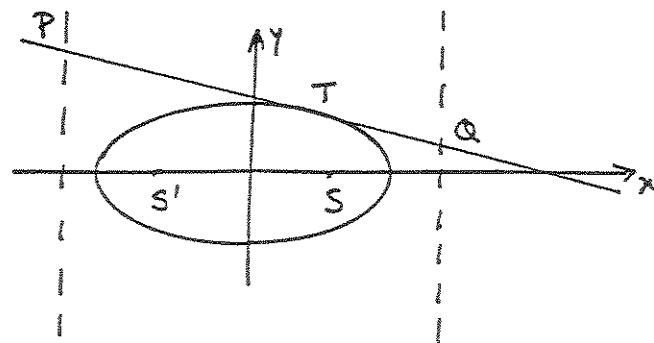
(b)



In the diagram at left, the chords PQ and CD are parallel.
The tangent at D cuts the chord PQ at T.
The other point of contact from T is B and BC cuts PQ at R.

- | | |
|---|---|
| | (i) Copy the diagram onto your page |
| 3 | (ii) Prove that $\angle BDT = \angle BRT$ and state why B, T, D and R are concyclic |
| 3 | (iii) Show that $\triangle RCD$ is isosceles |

(c)



The tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $T(a \cos \theta, b \sin \theta)$ meets the directrices of the ellipse at P and Q.

S and S' are the foci.

Show that $\angle TSQ = 90^\circ$

QUESTION 5:

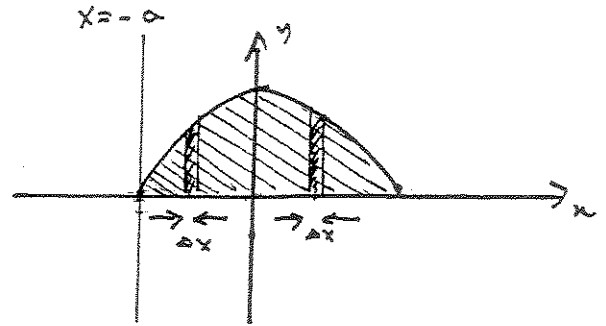
- (a) Sketch, on separate axes, the following graphs, showing all important features
(DO NOT use Calculus)
- 2 (i) $y = \sin^2 x \quad -2\pi \leq x \leq 2\pi$
- 2 (ii) $y = \ln\left(\frac{1}{x}\right) \quad x > 0$
- 2 (iii) $y = \frac{\sin x}{x} \quad x > 0$
- 2 (iv) $y = \max(x, 1-x)$ where $\max(a, b) = a$ when $a \geq b$
 $ \phantom{a < b}$
 $ \phantom{a < b}$
- 2 (b) (i) Use De Moivre's Theorem to show that
- $$(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = \frac{2 \cos n\theta}{\cos^n \theta} \quad \text{where } n \text{ is an integer}$$
- ($\cos \theta \neq 0$)
- 3 (ii) Use this result to show that the equation
- $$(1+z)^4 + (1-z)^4 = 0 \quad \text{has roots of } \pm i \tan \frac{\pi}{8}, \pm i \tan \frac{3\pi}{8}$$
- 2 (iii) Hence, or otherwise, show that $\tan^2 \frac{\pi}{8} = 3 - 2\sqrt{2}$

QUESTION 6:

5 (a) Find $\int_0^1 \sqrt{4 - (1+x)^2} dx$

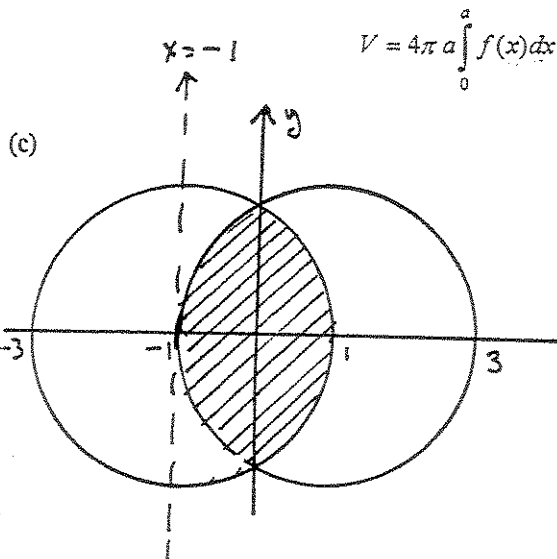
(b) The curve $y=f(x)$ is reflected in the y -axis to give the shape shown

The strips shown both have width Δx and are equidistant from the y -axis.



3 (i) The shaded area is rotated around the line ~~$x = -a$~~ $x = -a$. Find each of the volumes of the two cylindrical shells as the two strips are rotated. (Δx is small)

3 (ii) Show that the volume of the solid so formed is given by



Two circles, centres $(-1,0)$ and $(1,0)$ and of radii 2 units have a common region as shown, and this region is rotated about ~~$x = -1$~~ $x = -1$

2 (i) Show that the volume of the solid formed is given by

$$V = 8\pi \int_0^1 \sqrt{4 - (x+1)^2} dx$$

2 (ii) By using your answer to part (a) of this question above, find the exact volume of the solid.

QUESTION 7:

- (a) A particle moves in a straight line so that its distance from the origin at any time t is given by x and its velocity by v .

- 3 (i) The acceleration of the particle at a distance x is given by the equation

$$a = n^2(3 - x) \quad \text{where } n \text{ is a constant.}$$

If the particle moves from rest from the origin ($x=0$), show that

$$\frac{1}{2}v^2 - n^2\left(3x - \frac{1}{2}x^2\right) = 0$$

- 2 (ii) Hence show that the particle never moves outside a certain interval and give that interval.

- 5 (b) (i) Let $I_n = \int_1^e x(\ln x)^n dx$ where $n=0,1,2,3,\dots$

Using integration by parts, show that

$$I_n = \frac{e^2}{2} - \frac{n}{2}I_{n-1} \quad n=1,2,3,\dots$$

- 5 (ii) The area bounded by the curve $y = \sqrt{x}(\ln x)$ $x \geq 1$

the x -axis and the line $x=e$ is rotated about the x -axis through 2π radians.

Find the exact value of the volume of the solid of revolution so formed.

QUESTION 8:

4 (a)

Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos x + \sin x}$ using the substitution $t = \tan \frac{x}{2}$

4 (b)

A plane curve is defined by $x^2 + 2xy + y^5 = 4$

This curve has a horizontal tangent at the point $P(X, Y)$

By using Implicit Differentiation (or otherwise), show that X is the unique real root of

$$X^5 + X^2 + 4 = 0$$

3

(c) (i)

If $x_1 > 1$ and $x_2 > 1$ show that $x_1 + x_2 > \sqrt{x_1 x_2}$

4

(ii)

Use the Principle of Mathematical Induction to show that, for $n \geq 2$, if $x_j > 1$ where $j=1, 2, 3, \dots, n$ then

$$\ln(x_1 + x_2 + \dots + x_n) > \frac{1}{2^{n-1}} (\ln x_1 + \ln x_2 + \dots + \ln x_n)$$

QUESTION 1

a) i) $\int \frac{dx}{(x+1)^2+4} = \frac{1}{2} \tan^{-1} \frac{x+1}{2} + k$ ✓

ii) $\int_0^1 (x+1)^{-\frac{3}{2}} dx = \left[-2(x+1)^{-\frac{1}{2}} \right]_0^1$ ✓
 $= 2 - \sqrt{2}$ ✓

b) $\frac{\sec x \tan x + \sec^2 x}{\sec x + \tan x}$

$= \frac{\sec x (\tan x + \sec x)}{(\tan x + \sec x)} = \sec x$ ✓

$\int \sec x dx = \int \frac{\sec x \tan x + \sec^2 x}{\sec x + \tan x} dx$ ✓
 $= \ln(\sec x + \tan x) + k.$ ✓

c) $\int_0^1 x e^{-x} dx = \left[-x e^{-x} \right]_0^1 + \int_0^1 e^{-x} dx$ ✓

i) $= \left[-x e^{-x} - e^{-x} \right]_0^1$ ✓
 $= \left[-\frac{1}{e} - \frac{1}{e} \right] - [0 - 1]$ ✓
 $= 1 - \frac{2}{e}$ ✓

ii) $\frac{5}{(x+1)(x^2+4)} = \frac{a}{x+1} + \frac{bx+c}{x^2+4}$
 $= \frac{ax^2+4a+bx^2+cx+bx+c}{(x+1)(x^2+4)}$

$\therefore 4a+c=5$
 $\begin{cases} a+b=0 \\ b+c=0 \end{cases} \Rightarrow \begin{cases} a=-b \\ c=-b \end{cases}$
 $\therefore a=c$

$\therefore 4a+a=5$

$\therefore a=1$

$\therefore c=1$

$\therefore b=-1$

i) All students OK on this

Many students unable to do the arithmetic to get correct answer. Need to write it out in detail - not carry signs in their head.

Cancel common factor

Numerator is derivative of denominator

• must put terminals on $-x e^{-x}$

• Again, many students lost track of minus signs here

← Set it out properly

← Setting up correct numerators is basic (but important)

Q1 c) (ii) (cont)

$$\begin{aligned}\therefore \int \frac{5dx}{(x+1)(x^2+4)} &= \int \frac{1}{x+1} + \frac{1-x}{(x^2+4)} dx \quad \checkmark \\ &= \int \frac{dx}{x+1} + \int \frac{dx}{x^2+4} - \frac{1}{2} \int \frac{2x}{x^2+4} dx \\ &= \ln(x+1) + \frac{1}{2} \tan^{-1} \frac{x}{2} - \frac{1}{2} \ln(x^2+4) + c\end{aligned}$$

$$\begin{aligned}1) \int_k^1 \frac{dx}{x(x+1)} &= \int_k^1 \frac{1}{x} - \frac{1}{x+1} dx \quad \checkmark \\ &= \left[\ln x - \ln(x+1) \right]_k^1 \\ &= (\ln 1 - \ln 2) - (\ln k - \ln(k+1)) \\ &= \ln(k+1) - \ln k - \ln 2 \quad \checkmark\end{aligned}$$

$$\begin{aligned}\text{Now } \sum_{k=1}^n \int_k^1 \frac{dx}{x(x+1)} &= \sum_{k=1}^n (\ln \frac{k+1}{k} - \ln 2) \\ &= \ln \frac{2}{1} - \ln 2 + \ln \frac{3}{2} - \ln 2 \\ &\quad + \dots + \ln \frac{n}{n-1} - \ln 2 + \ln \frac{n+1}{n} - \ln 2 \\ &= (\ln 2 + \ln \frac{3}{2} + \dots + \ln \frac{n}{n-1} + \ln \frac{n+1}{n}) - n \ln 2 \\ &= \ln (2 \cdot \frac{3}{2} \cdot \frac{4}{3} \cdot \dots \cdot \frac{n}{n-1} \cdot \frac{n+1}{n}) - n \ln 2 \\ &= \ln(n+1) - n \ln 2\end{aligned}$$

This question generally handled well

Most students could generate these partial fractions easily.

Many forms of answer possible eg. $\left[\ln \frac{(k+1)}{k} - \ln 2 \right]$ Hint: look below at next part of question & leave $\ln 2$ separate.

*Write out a few terms of the series 1st, 2nd ... penultimate, last. Look for the pattern.

Intermediate steps must be shown

Show cancelling

No marks for last line

QUESTION 2:

(a) $z = 3 - 4i$

(i) $\bar{z} = 3 + 4i$ (ii) $|z| = 5$ (iii) $\arg z = \tan^{-1}(-4/3)$
 $= -53^\circ 8'$

(iv) Let $a + ib = \sqrt{3 - 4i}$

$$a^2 - b^2 = 3$$

$$2ab = -4$$

$$b = -\frac{2}{a}$$

$$a^2 - \frac{4}{a^2} = 3$$

$$a^4 - 4 = 3a^2$$

$$(a^2 + 1)(a^2 - 4) = 0$$

$$a = \pm 2 \text{ or } a = \pm i$$

$$\therefore b = \mp 1$$

$$\therefore \sqrt{z} = \pm(2 - i)$$

(b) $\sqrt{x^2 + (y-1)^2} = y$

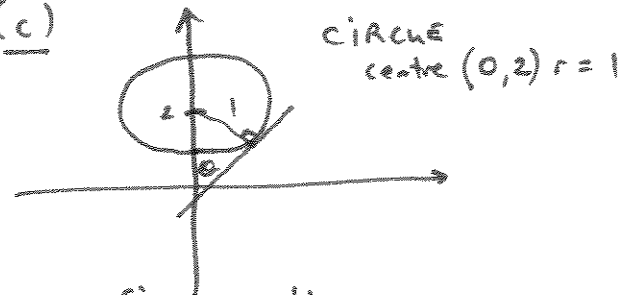
$$x^2 + y^2 - 2y + 1 = y^2$$

$$x^2 - 2y + 1 = 0$$

$$y = \frac{1}{2}(x^2 + 1)$$

A parabola, vertex $(0, \frac{1}{2})$

(c)



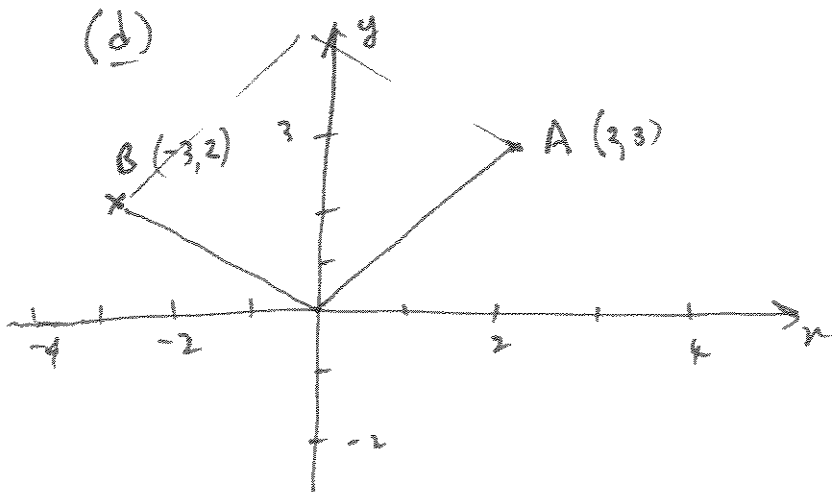
$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}$$

$$\therefore \text{Smallest angle} = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$$

note that the tangent to the circle gives the most extreme point of the circle. i.e. the least value of $\arg z$ is the angle the tangent makes with x-axis

(d)



By inspection C is $(-1, 5)$

QUESTION 3.

(a) $x^3 + ax^2 + bx + c = 0$

Let the roots be $\alpha, \beta, \alpha + \beta$.

Sum $\alpha + \beta + \alpha + \beta = -a$
 $2\alpha + 2\beta = -a/2$

Product (x2) $\alpha\beta + \alpha(\alpha + \beta) + \beta(\alpha + \beta) = b$

$\therefore \alpha\beta + (\alpha + \beta)(\alpha + \beta) = b$

$\therefore \alpha\beta + a^2/4 = b \Rightarrow \alpha\beta = b - a^2/4$

Product $\alpha\beta(\alpha + \beta) = -c$

$\therefore (b - a^2/4)(-a/2) = -c$

$\therefore -ab/2 + a^3/8 = -c$

$\therefore a^3 - 4ab + 8c = 0$

ALTERNATIVELY

$\rightarrow \therefore \gamma = -a/2$

Now γ is a root so

$P(\gamma) = 0$

$\therefore (-a/2)^3 + a(-a/2)^2 + b(-a/2) = 0$

$\therefore -a^3/8 + a^3/4 - ab/2 + c = 0$

$\therefore -a^3 + 2a^3 - 4ab + 8c = 0$

$\therefore a^3 - 4ab + 8c = 0$

(b) $P(x) = x^3 + ax^2 + bx + 6$

$1-i$ is a root.

$\therefore$ So is $1+i$

$\therefore (x - (1-i))(x - (1+i))$ is a factor

$\therefore (x-1+i)(x-1-i)$

ie. $(x-1)^2 - i^2$

$\therefore P(x) = x^3 + ax^2 + bx + 6 = (x^2 - 2x + 2)Q(x)$

By inspection $Q(x) = (x+3)$

and so $a = +1$

$b = -4$

and $P(x) = (x^2 - 2x + 2)(x+3)$

ALTERNATIVELY

Product of roots

$\Rightarrow (1-i)(1+i)\alpha = -6$

$\therefore \alpha = -3$ (*)

Sum of Roots = $-a$

$\therefore a = 1$

Product (x2)

$= 2 - 3 + 3i - 3 - 3i = 2$

$b = -4$

Result (*) gives $P(x) = (x+3)Q(x)$

division giving $Q(x)$ as $x^2 - 2x + 2$

ie. $P(x) = (x^2 - 2x + 2)(x+3)$

Some very "shoddy" proofs here.
 The most popular was to find $P(1-i)$ and $P(1+i)$ and solve simultaneously. The other was to perform a long division using $x-1+i$.
 These methods show little appreciation of Polynomial Theory outside the 2/3 unit factor theorems. Chances are, in 4 unit, we are always going to use Sum of Roots, etc...

(c) (i) $y = 1/x$ and $y = x^2 - 4$

$1/x = x^2 - 4$

$x^3 - 4x - 1 = 0$

Comment: Easy mark

(ii) In case, $\alpha + \beta + \gamma = 0$
 $\alpha\beta + \beta\gamma + \alpha\gamma = -4$
 $\alpha\beta\gamma = 1$

Sum of new roots $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$
 $= 0 + 8 = 8$

Product of new roots $\alpha^2\beta + \alpha^2\gamma + \beta^2\gamma = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$
 $= 16 - 2 \cdot 1 \cdot 0$
 $= 16$

Product of new roots $\alpha^2\beta^2\gamma^2 = 1$

New polynomial is $x^3 - 8x^2 + 16x - 1 = 0^*$

ALTERNATIVELY (if you know it)

Let $y = x^2$ $\therefore \sqrt{y} = x$

$\therefore x^3 - 4x - 1 = 0$ becomes

$(\sqrt{y})^3 - 4(\sqrt{y}) - 1 = 0$

ie. $y^{3/2} - 4y^{1/2} - 1 = 0$

Squaring both sides gives,

$y^3 + 16y - 8y^2 = 1$

ie. $y^3 - 8y^2 + 16y - 1 = 0$

(iii) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{\beta^2\gamma^2 + \alpha^2\gamma^2 + \alpha^2\beta^2}{\alpha^2\beta^2\gamma^2}$

$= 16/1$ if you used method I above.

OR for the polynomial marked (*) above,

$= \frac{\text{Sum of roots taken 2 at a time}}{\text{Product of roots}}$

$= \frac{+16}{-(-1)} = 16$

This is the most complicated method but the easiest to understand. When in doubt, use this case than half-braining a formula.

A lot of people "half" know this method. A lot left the polynomial as this line, but this is not a polynomial (powers of x are not integral)

(iv) $OP^2 = (2-0)^2 + (\frac{1}{2}-0)^2$ by distance formula
 $= 2^2 + \frac{1}{4}^2$

Similarly, $OQ^2 = \beta^2 + \frac{1}{\beta^2}$ and $OR^2 = \gamma^2 + \frac{1}{\gamma^2}$.

Using previous 2 answers

$$OP^2 + OQ^2 + OR^2 = 2^2 + \beta^2 + \gamma^2 + \left(\frac{1}{2^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}\right)$$
$$= 8 + 16 = 24$$

Question 4:

(a) $9x^2 - 16y^2 = 144$
 $\frac{x^2}{16} - \frac{y^2}{9} = 1$

$$a = 4 \quad b = 3.$$

- (i) 8 (ii) $\frac{5}{4}$ (iii) Foci are $(\pm 5, 0)$ (iv) $\mathcal{Q}$: $x = \pm \frac{16}{5}$
(v) Asymptotes are $y = \pm \frac{3x}{4}$



- (i) Let $\angle BDT = x^\circ$
 $\therefore \angle BCD = x^\circ$ (angle in the alternate segment is the same as angle made by tangent striking a chord DB)
 $\angle BCD = \angle BAT = x^\circ$ (corresponding angles, $PT \parallel CD$)
- (ii) Since $\angle BRT = \angle BDT$ they can be considered as angles standing on arc BT. i.e. Circle goes through B, T, D, R
- (iii) $\angle TBD = x^\circ$ (tangents striking a circle make the same angle with the chord of contact) OR (use alt seg theorem with $\angle BCD$)
 $\angle TBD = \angle TRD = x^\circ$ (angles on circumference on arc TD of circle touching B, T, D, R)
 $\angle TRD = \angle RDC = x^\circ$ (alternate angles, $PT \parallel CD$)
 $\therefore \angle BCD = \angle RDC = x^\circ$
 $\therefore \triangle RCD$ is isosceles

$$(c) \quad \frac{dy}{dx} = -\frac{2n}{a^2} \times \frac{b^2}{2y}$$

$$= -\frac{b^2}{a^2} \cdot \frac{n}{y}$$

At T ($a \cos \theta, b \sin \theta$)

$$m_T = -\frac{b^2}{a^2} \frac{a \cos \theta}{b \sin \theta}$$

$$= -\frac{b \cos \theta}{a \sin \theta}$$

Tangent is:

$$y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$\therefore a y \sin \theta - a b \sin^2 \theta = -b x \cos \theta + a b \cos^2 \theta$$

$$\therefore a y \sin \theta + b x \cos \theta = a b$$

$$\text{or } \frac{y \sin \theta}{b} + \frac{x \cos \theta}{a} = 1$$

At Q $x = + \frac{a}{e}$

$$\therefore \frac{y \sin \theta}{b} + \frac{a \cos \theta}{a} = 1$$

$$\therefore \frac{y \sin \theta}{b} = 1 - \frac{a \cos \theta}{a}$$

$$\therefore \frac{y \sin \theta}{b} = \frac{a b - \frac{a b \cos \theta}{e}}{a}$$

$$= \frac{b(e - \cos \theta)}{e}$$

$$\therefore y = \frac{b(e - \cos \theta)}{e \sin \theta} \quad \therefore Q \text{ is } \left(\frac{a}{e}, \frac{b(e - \cos \theta)}{e \sin \theta} \right)$$

$$m_{TS} = \frac{b \sin \theta}{a \cos \theta - a e}$$

$$m_{SQ} = \frac{b(e - \cos \theta)}{e \sin \theta} \bigg/ \frac{a}{e} - a e$$

$$= \frac{e b(e - \cos \theta)}{e \sin \theta} \bigg/ a - a e^2$$

$$= \frac{b(e - \cos \theta)}{\sin \theta} \bigg/ a - a e^2$$

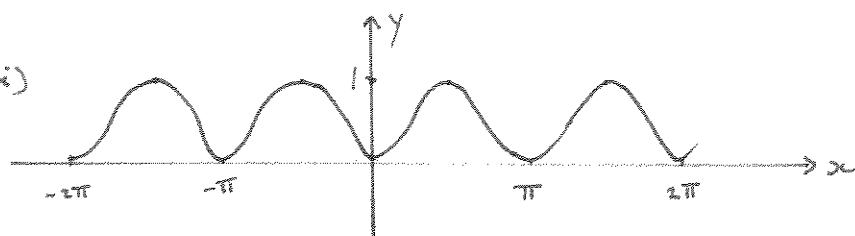
(multiplying top & bottom of $\frac{e}{e}$ by e)

$$m_{TS} \times m_{SQ} = \frac{b \sin \theta}{a(\cos \theta - e)} \times \frac{b(e - \cos \theta)}{\sin \theta} \times \frac{1}{a - a e^2}$$

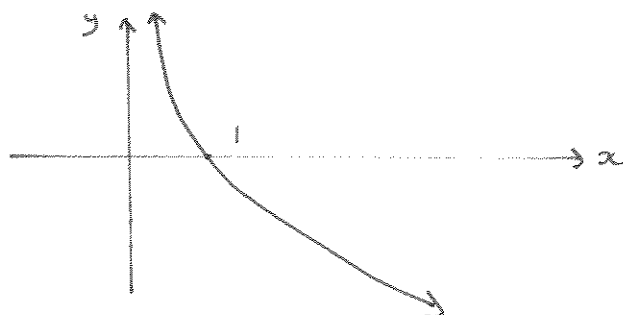
$$= \frac{b^2 (e - \cos \theta)}{a^2 (\cos \theta - e)(1 - e^2)} = -1 \quad \text{because } a^2(1 - e^2) = b^2$$

QUESTION 5

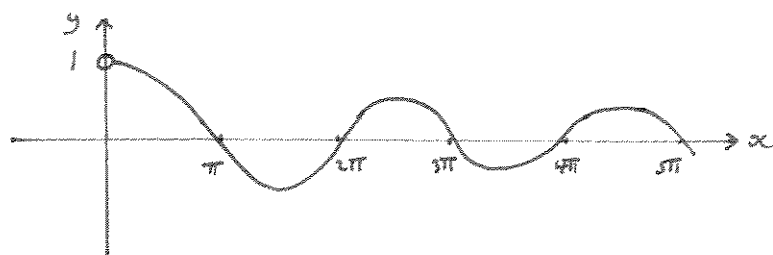
a) i)



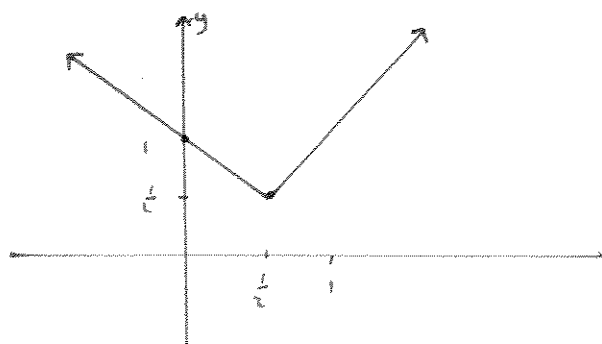
ii)



iii)



iv)



$$\begin{aligned}
 \text{b) i) LHS} &= (1 + i \tan \theta)^n + (1 - i \tan \theta)^n \\
 &= \left(\frac{\cos \theta + i \sin \theta}{\cos \theta} \right)^n + \left(\frac{\cos \theta - i \sin \theta}{\cos \theta} \right)^n \\
 &= \frac{\cos^n \theta + i \sin^n \theta + \cos^n \theta - i \sin^n \theta}{\cos^n \theta} \\
 &= \frac{2 \cos^n \theta}{\cos^n \theta} \\
 &= \text{RHS}
 \end{aligned}$$

question clearly said to show all important features - x intercepts, etc.

curves should be smooth except for sharp corner in (iv).

make sure each step clearly follows on from previous step.

$$ii) (1+z)^4 + (1-z)^4 = 0$$

$$\text{let } z = i \tan \theta$$

$$\therefore (1+i \tan \theta)^4 + (1-i \tan \theta)^4 = 0$$

$$\therefore \frac{2 \cos 4\theta}{\cos^4 \theta} = 0 \quad \text{from (i)}$$

$$\therefore \cos 4\theta = 0$$

$$\therefore 4\theta = \frac{\pi}{2}, \frac{3\pi}{2}, -\frac{\pi}{2}, -\frac{3\pi}{2}, \dots$$

$$\theta = \pm \frac{\pi}{8}, \pm \frac{3\pi}{8}$$

$$\therefore z = i \tan\left(\pm \frac{\pi}{8}\right), i \tan\left(\pm \frac{3\pi}{8}\right)$$

$$z = \pm i \tan \frac{\pi}{8}, \pm i \tan \frac{3\pi}{8}$$

$$iii) (1+z)^4 + (1-z)^4 = 0$$

$$1 + 4z + 6z^2 + 4z^3 + z^4 + 1 - 4z + 6z^2 - 4z^3 + z^4 = 0$$

$$2 + 12z^2 + 2z^4 = 0$$

$$z^4 + 6z^2 + 1 = 0$$

$$z^2 = \frac{-6 \pm \sqrt{32}}{2}$$

$$= -3 \pm \sqrt{8}$$

$$\therefore \left(\pm i \tan \frac{\pi}{8}\right)^2 = -3 + \sqrt{8} \quad \text{from part ii)}$$

$$-\tan^2 \frac{\pi}{8} = -3 + 2\sqrt{2}$$

$$\tan^2 \frac{\pi}{8} = 3 - 2\sqrt{2}$$

very few got this far

QUESTION 6:

(a) $\int \sqrt{4 - (1+x)^2} dx$

Let $1+x = 2\sin\theta$ $\begin{cases} x=0 & \theta = \pi/6 \\ x=1 & \theta = \pi/2 \end{cases}$
 $\frac{dx}{d\theta} = 2\cos\theta$
 $\Rightarrow dx = 2\cos\theta d\theta$
 $\int_0^1 \sqrt{4 - (1+x)^2} dx = \int_{\pi/6}^{\pi/2} 2\sqrt{1 - \sin^2\theta} 2\cos\theta d\theta$
 $= 4 \int_{\pi/6}^{\pi/2} \cos\theta \cos\theta d\theta$
 $= 4 \int_{\pi/6}^{\pi/2} \frac{\cos 2\theta + 1}{2} d\theta$
 $= 2 \left[\frac{1}{2} \sin 2\theta + \theta \right]_{\pi/6}^{\pi/2}$
 $= \left(\sin \pi + \frac{\pi}{2} \right) - \left(\sin \frac{\pi}{3} + \frac{\pi}{3} \right)$
 $= 2\pi/3 - \frac{\sqrt{3}}{2}$

If you can do all sorts of integration (and nothing else) you can go very close to passing 4 UNITS.

(b) 
 $VOL_{L1} = 2\pi(a-x)y \Delta x$


 $VOL_2 = 2\pi(a+x)y \Delta x$

Note: x is a variable!
 In these 2 cases x is different and will trace over different limits.
 The question asked for each of the volumes.

(ii) VOL of the RHS = $2\pi \int_0^a (a+x)y dx$

VOL of the LHS = $2\pi \int_{-a}^0 (a-x)y dx$

(changing limits around) $= 2\pi \int_0^a (a-x)y dx$

This is true because $f(x)$ is an even function $= 2\pi \int_0^a (a-x)y dx$

$\therefore VOL = 2\pi \int_0^a ((a-x) + (a+x))y dx$
 $= 4\pi \int_0^a xy dx$

NOTE the different limits here. Most students now just added these 2 integrals and (surprisingly) ignored the limits. You need to explain how this can be done.

(c)⁽ⁱ⁾ Comparing this diagram with the first.

$$\begin{aligned} - f(x) &= \sqrt{4 - (x+1)^2} & \left\{ \begin{array}{l} \text{since } y^2 + (x+1)^2 = 4 \\ \text{and we are using } f(x) \text{ as the} \\ \text{RHS in the first diagram} \end{array} \right. \\ - a &= 1 \end{aligned}$$

- The second diagram will have twice the volume of the first (due to part below the x-axis)

not many students saw this subtle point and "judged" the sum of the integrals.

From part (b),

$$\begin{aligned} V &= 2 \times \left[4\pi(1) \int_0^1 \sqrt{4 - (x+1)^2} dx \right] \\ &= 8\pi \int_0^1 \sqrt{4 - (x+1)^2} dx \end{aligned}$$

a lot of people proved the whole formula again here. Use part (b).

(ii) From part (a), $\int_0^1 \sqrt{4 - (x+1)^2} dx$

$$= 2\pi/3 - \frac{\sqrt{3}}{2}$$

$$\begin{aligned} \therefore \text{VOL} &= 8\pi \left(2\pi/3 - \frac{\sqrt{3}}{2} \right) \\ &= \frac{16\pi^2}{3} - 4\pi\sqrt{3} \end{aligned}$$

QUESTION 7

a) i) $a = n^2(3-x)$

$$\therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = n^2(3-x)$$

$$\frac{1}{2} v^2 = n^2 \left(3x - \frac{1}{2} x^2 + c \right)$$

when $x=0$ $v=0 \Rightarrow c=0$

$$\therefore \frac{1}{2} v^2 = n^2 \left(3x - \frac{1}{2} x^2 \right)$$

$$\therefore \frac{1}{2} v^2 - n^2 \left(3x - \frac{1}{2} x^2 \right) = 0$$

ii) $3x - \frac{1}{2} x^2 \geq 0$ as $\frac{1}{2} v^2 \geq 0$

$$6x - x^2 \geq 0$$

$$0 \leq x \leq 6$$

b) i) $I_n = \int_1^e x (\ln x)^n dx$ $u = (\ln x)^n$ $v = \frac{1}{2} x^2$
 $u' = \frac{n(\ln x)^{n-1}}{x}$ $v' = x$

$$= \left[\frac{1}{2} x^2 (\ln x)^n \right]_1^e - \frac{1}{2} \int_1^e x^2 \cdot \frac{n(\ln x)^{n-1}}{x} dx$$

$$= \left[\frac{e^2}{2} - 0 \right] - \frac{n}{2} \int_1^e x (\ln x)^{n-1} dx$$

$$= \frac{e^2}{2} - \frac{n}{2} I_{n-1}$$

ii) Volume = $\pi \int_1^e x (\ln x)^2 dx$

$$= \pi \left(\frac{e^2}{2} - I_1 \right)$$

$$= \pi \left(\frac{e^2}{2} - \left(\frac{e^2}{2} - \frac{1}{2} I_0 \right) \right)$$

$$= \pi \left(\frac{e^2}{2} - \left(\frac{e^2}{2} - \frac{1}{2} \left(\frac{e^2}{2} - \frac{1}{2} \right) \right) \right)$$

$$= \frac{\pi}{4} (e^2 - 1) \quad \text{cubic units}$$

need to evaluate $c=0$.

limits and dx should be written where appropriate

a lot of careless errors here.

QUESTION 8

2) $\int_0^{\pi/2} \frac{dx}{1 + \cos x + \sin x}$ where $t = \tan \frac{x}{2}$

Now $dx = \frac{2dt}{1+t^2}$ *

$\cos x = \frac{1-t^2}{1+t^2}$

$\sin x = \frac{2t}{1+t^2}$

Integral becomes $\int_0^1 \frac{\frac{2dt}{1+t^2}}{1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}}$ ✓

$= \int_0^1 \frac{\frac{2dt}{1+t^2} \times (1+t^2)}{\left(1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}\right) \times (1+t^2)}$

$= \int_0^1 \frac{2dt}{1+t^2 + 1-t^2 + 2t}$

$= \int_0^1 \frac{2dt}{2+2t}$

$= \int_0^1 \frac{dt}{1+t} = \left[\ln(1+t) \right]_0^1$
 $= \ln 2$ ✓

1) Now $x^2 + 2xy + y^5 = 4$

$\frac{d}{dx}(x^2) + \frac{d}{dx}(2xy) + \frac{d}{dx}(y^5) = \frac{d}{dx}(4)$

i.e. $2x + 2y + 2x \frac{dy}{dx} + 5y^4 \frac{dy}{dx} = 0$ ✓

$\therefore \frac{dy}{dx} = \frac{-2(x+y)}{2x+5y^4}$
 $= 0$ when $x = X, y = Y$ ✓

LEARN THESE (incl. $\frac{dx}{dx}$)

to save time.

Learn how to simplify compound fractions. Hint: best method is to multiply top & bottom by highest denominator. $(1+t^2)$ in this case

Spend time to do this step. This is implicit differentiation ←

Use product rule for $2xy$. Don't forget $RHS = 0$

Horizontal tangent (stat. point)

$$\therefore -2(X+Y) = 0$$

$$\therefore X = -Y$$

$$\text{or } Y = -X$$

Substituting into original equation gives

$$X^2 + 2X(-X) + (-X)^5 = 4$$

$$\text{ie } X^2 - 2X^2 - X^5 = 4$$

$$\text{or } X^2 + X^5 + 4 = 0 \text{ as reqd.}$$

$$\text{c) (i) } x_1 + x_2 > \sqrt{x_1 x_2}$$

$$\text{Now } (\sqrt{x_1} - \sqrt{x_2})^2 > 0$$

$$\text{ie } x_1 - 2\sqrt{x_1 x_2} + x_2 > 0$$

$$\text{ie } x_1 + x_2 > 2\sqrt{x_1 x_2}$$

$$\therefore x_1 + x_2 > \sqrt{x_1 x_2}$$

OR

Now if $x_1 + x_2 > \sqrt{x_1 x_2}$, by squaring both sides we obtain

$$x_1^2 + 2x_1 x_2 + x_2^2 > x_1 x_2$$

$$\text{ie } x_1^2 + x_1 x_2 + x_2^2 > 0 \text{ which}$$

must be true since both x_1, x_2 are greater than 1.

$\therefore$ the original statement

$$x_1 + x_2 > \sqrt{x_1 x_2} \text{ must have}$$

been true.

$$\text{i) To prove } \ln(x_1 + x_2 \cdots x_n) > \frac{1}{2^{n-1}} (\ln x_1 + \ln x_2 + \cdots + \ln x_n)$$

for $n \geq 2$.

Now when $n=2$, we know

$$x_1 + x_2 > \sqrt{x_1 x_2} \text{ from above}$$

$$\therefore \ln(x_1 + x_2) > \ln \sqrt{x_1 x_2} \text{ since}$$

$x_1 + x_2$ & $x_1 x_2$ are both > 1 .

NEVER START WITH THE STATEMENT

YOU ARE REQUIRED TO PROVE

You cannot get full marks by working on the statement you have to prove (unless you are very clever).

* Use the word "Now" to signal the 1st line of your argument. (ALWAYS!)

* use "ie." to write the same thing but in a different form

* " $\therefore$ " means something new based on what came before.

* If you try to do the 2nd version of this proof and don't use the words to explain your reasoning you WILL LOSE MARKS!

$$\text{i.e. } \ln(x_1 + x_2) > \ln(x_1 x_2)^{\frac{1}{2}}$$

$$\text{i.e. } \ln(x_1 + x_2) > \frac{1}{2} \ln(x_1 x_2)$$

$$\text{i.e. } \ln(x_1 + x_2) > \frac{1}{2} (\ln x_1 + \ln x_2)$$

as reqd

$\therefore$ true for $n=2$. ✓

Assume statement true when $n=k$

$$\text{i.e. } \ln(x_1 + x_2 + \dots + x_k) > \frac{1}{2^{k-1}} (\ln x_1 + \dots + \ln x_k)$$

when $n = k+1$,

$$\text{LHS} = \ln(x_1 + x_2 + \dots + x_k + x_{k+1})$$

$$> \frac{1}{2} (\ln(x_1 + x_2 + \dots + x_k) + \ln x_{k+1})$$

from result proved for $n=2$.

$$> \frac{1}{2} \left(\frac{1}{2^{k-1}} (\ln x_1 + \ln x_2 + \dots + \ln x_k) + \ln x_{k+1} \right)$$

$$> \frac{1}{2} \left(\text{ " } + \frac{1}{2^{k-1}} \ln x_{k+1} \right)$$

since $\frac{1}{2^{k-1}} < 1$ ✓

$$= \frac{1}{2^k} (\ln x_1 + \ln x_2 + \dots + \ln x_k + \ln x_{k+1})$$

which is correct form for RHS when $n = k+1$.

$\therefore$ By theory of Mathematical Induction the statement is true for all $n \geq 2$.

Proof for $n=2$ must refer back to statement proved in (i)

Do NOT WRITE THE STATEMENT YOU'RE TRYING TO PROVE AND THEN JUST WORK ON IT. MUST USE LHS = RHS = etc.

← using the assumption. This must be used somewhere in your proof.

* Many students invented their own log laws!

$$\text{eg } \ln(x_1 + x_2 + x_3) = \ln(x_1 + x_2) \cdot \ln x_3 \quad (\text{try with } x_3 = 1)$$

* Note correct use of $=$ & $>$, each refers to line above.

* Don't waste time in a lengthy conclusion. No marks for it (usually)!